

OOPs assignment 5

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Batch:2o13

Q1 Write a simple base class, then a derived class and use objects of both of them in the main function. It will be a simple illustration of inheritance

```
ain.cpp
1  #include <iostream>
2  using namespace std;
3
4  class Base {
5  public:
6      void show() {
7          cout << "Base class\n";
8      }
9  };
10
11 class Derived : public Base {
12 public:
13     void display() {
14         cout << "Derived class\n";
15     }
16 };
17
18 int main() {
19     Base b;
20     Derived d;
21
22     b.show();
23     d.show();
24     d.display();
25
26     return 0;
27 }
```

```
Base class
Base class
Derived class

...Program finished with exit code 0
Press ENTER to exit console.
```

Q2 Practice protected access specifier in inheritance. In the base class declare a variable which is protected and access it in the derived class.

```
main.cpp
1  #include <iostream>
2  using namespace std;
3
4  class Base {
5  protected:
6      int x;
7  public:
8      void set() {
9          x = 10;
10     }
11 };
12
13 class Derived : public Base {
14 public:
15     void show() {
16         cout << x;
17     }
18 };
19
20 int main() {
21     Derived d;
22     d.set();
23     d.show();
24     return 0;
25 }
```

input

```
10
...Program finished with exit code 0
Press ENTER to exit console.
```

Q3 Most of the time we use public mode of inheritance, for example *class Derived: public Base{}*; Try protected and private access modifiers to understand the difference of various modes of inheritance.

```
main.cpp
1  #include <iostream>
2  using namespace std;
3
4  class Base {
5  public:
6      int a = 1;
7  protected:
8      int b = 2;
9  private:
10     int c = 3;
11 };
12
13 class PublicDerived : public Base {
14 public:
15     void show() {
16         cout << a << " " << b;
17     }
18 };
19
20 class ProtectedDerived : protected Base {
21 public:
22     void show() {
23         cout << a << " " << b;
24     }
25 };
26
27 class PrivateDerived : private Base {
28 public:
```

```
24 }
25 };
26
27 class PrivateDerived : private Base {
28 public:
29     void show() {
30         cout << a << " " << b;
31     }
32 };
33
34 int main() {
35     PublicDerived p;
36     ProtectedDerived pr;
37     PrivateDerived pv;
38
39     p.show();
40     pr.show();
41     pv.show();
42
43     return 0;
44 }
```

input

```
1 21 21 2
...Program finished with exit code 0
Press ENTER to exit console.
```

input

Q4 Various types of inheritances are as shown below. Write small C++ codes for each inheritance type.

main.cpp

```
1  #include <iostream>
2  using namespace std;
3
4  class A {
5  public:
6      void showA() { cout << "A\n"; }
7  };
8
9  class B : public A {};
10 class C : public A {};
11
12 class D : public B, public C {
13 };
14
15 int main() {
16     D d;
17     d.B::showA();
18     d.C::showA();
19 }
```

input

```
A
A

...Program finished with exit code 0
Press ENTER to exit console.
```

Q5 How can you use constructors and destructors in C++ inheritance? Write a program to illustrate.

```
main.cpp
1  #include <iostream>
2  using namespace std;
3
4  class Base {
5  public:
6      Base() { cout << "Base Constructor\n"; }
7      ~Base() { cout << "Base Destructor\n"; }
8  };
9
10 class Derived : public Base {
11 public:
12     Derived() { cout << "Derived Constructor\n"; }
13     ~Derived() { cout << "Derived Destructor\n"; }
14 };
15
16 int main() {
17     Derived d;
18     return 0;
19 }
```

```
input
Base Constructor
Derived Constructor
Derived Destructor
Base Destructor

...Program finished with exit code 0
Press ENTER to exit console.
```

Q6 Implement a base class Book with attributes title, author, and price. Then, create a derived class Textbook that inherits from Book and adds a new attribute subject. Demonstrate how single inheritance is used to manage the data for general books and textbooks.

```
ain.cpp
1  #include <iostream>
2  using namespace std;
3
4  class Book {
5  protected:
6      string title, author;
7      float price;
8  public:
9      void set(string t, string a, float p) {
10         title = t;
11         author = a;
12         price = p;
13     }
14 };
15
16 class Textbook : public Book {
17     string subject;
18 public:
19     void setData(string t, string a, float p, string s) {
20         set(t, a, p);
21         subject = s;
22     }
23     void show() {
24         cout << title << " " << author << " " << price << " " << subject;
25     }
26 };
27
28 int main() {
29     Textbook t;
30     t.setData("Math", "John", 500, "Algebra");
31 }

```

input

Math John 500 Algebra

...Program finished with exit code 0
Press ENTER to exit console.

Q7 A software company is creating a program to simulate a car's dashboard. The dashboard needs to display speed, fuel level, and temperature. The speed is controlled by a Speedometer class, the fuel level by a FuelGauge class, and the temperature by a Thermometer class. Implement the three classes: Speedometer, FuelGauge, and Thermometer, each with relevant attributes and methods. Then, create a CarDashboard class that inherits from all three classes to display the combined information on the dashboard. Demonstrate how multiple inheritance is used to build this class.

```

1  #include <iostream>
2  using namespace std;
3
4  class Speedometer {
5  protected:
6      int speed;
7  public:
8      void setSpeed(int s) { speed = s; }
9  };
10
11 class FuelGauge {
12 protected:
13     int fuel;
14 public:
15     void setFuel(int f) { fuel = f; }
16 };
17
18 class Thermometer {
19 protected:
20     int temp;
21 public:
22     void setTemp(int t) { temp = t; }
23 };
24
25 class CarDashboard : public Speedometer, public FuelGauge, public Thermometer
26 public:
27     void display() {
28         cout << speed << " " << fuel << " " << temp;
29     }
30 };

```

```

21 public:
22     void setTemp(int t) { temp = t; }
23 };
24
25 class CarDashboard : public Speedometer, public FuelGauge, public Thermometer
26 public:
27     void display() {
28         cout << speed << " " << fuel << " " << temp;
29     }
30 };
31
32 int main() {
33     CarDashboard c;
34     c.setSpeed(80);
35     c.setFuel(50);
36     c.setTemp(30);
37     c.display();
38 }

```




input

```
80 50 30

...Program finished with exit code 0
Press ENTER to exit console.
```